Edimah SYNESIUS SONGO

Applied mathematician · statistical reliability of AI evaluations and monitors

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</> Projects

False collusion alerts under incomplete information — October 2026

AI Collusion Research Sprint, Apart Research, solo, 23–25 October 2026

- In a simulated market, a standard screen's false-accusation rate rises 5.8% → 100% as auditor information degrades.
- A simple analytical model predicts these rates closely, without being fitted to them.
- Then proposed a "re-asking" test that bypasses the agents' secret channels.
- Exploratory pilot: 0/9 false flags, 3/3 planted pairs detected, with a statistic chosen after inspection.

Apart Research project page · github.com/Edimah/hackathon-ai-collusion

Randomisation Tests | Sobol Sensitivity Analysis | Simulation | Python

Detecting agentic collusion in public logs with statistical methods — September 2026

AI Incident Response Sprint, Apart Research × CeSIA, Track 2, solo

- Used statistical methods to detect an LLM-agent swarm from public edit logs alone.
- Reconstructed the logs from two public sources over 2003–2026.
- Designed a ladder of four null models to characterise the incident and its statistical significance.
- Built a weekly check as the incident-response deliverable.
- It flags the three largest incident weeks and none of the 2,236 earlier wiki-weeks (2016–2025).

Apart Research project page · github.com/Edimah/agentic-collusion-detector-in-public-logs

Change-Point Detection | Difference-in-Differences | Placebo Tests | Empirical p-values | Python

A Calibrated Two-Level Detector for Secret Loyalties – Application to an LLM — July 2026

"Secret Loyalties" hackathon, Apart Research × Formation Research, team of 2

- Designed an evaluation framework, then measured its false-positive rate on a clean model.
- Using matched-control calibration, cut false positives from 12/16 to 2/16.
- Result: top 25% of 179 submissions; one reviewer cited the calibration as "a key contribution".

Apart Research project page · github.com/edimah/secret-loyalties
Write-up: edimah.github.io/blog/2026/secret-loyalties-outputs-only/

Permutation Inference | Multiple-Comparison Correction

Advanced methods for model calibration with uncertain inputs — April – September 2023

M2 research thesis, EDF R&D, Chatou, solo

- Topic: Bayesian calibration of physical simulation models against noisy experimental data.
- Designed adaptive MCMC-based re-adjustment strategies, dubbed "MC x MCMC" and "quantisation x MCMC".
- Designed and implemented cut-distribution updates that respect the metrologists' calibration chain.

Bayesian Calibration | MCMC | Uncertainty Quantification | Python | OpenTURNS

Experience

Present – October 2025 — **Independent Consultant, Applied Statistics, AI Implementation**

- Statistics & Data consulting practice
- Flagship engagement: interim **Data Lead** at **CRCDC Guadeloupe** since February 2026.
- Rebuilding the regional cancer-screening data chain to deliver reliable data to stakeholders.
- Delivered a data-chain audit and methodology documentation under regulatory constraints.
- Assessed the feasibility of AI tools for pharmacy workflows (regulated health care).

Data Management | Communication | R | SQL | eSIS | DataQuery | Data Visualisation | Health Data Regulation

September 2024 – November 2023 — **Doctoral Researcher, IFP Energies Nouvelles / ONERA, Île-de-France**

- Multidisciplinary sensitivity analysis for complex physics systems.
- Development of sensitivity indices and explainability metrics in R and Python.
- Left the doctorate before completion.

R | Python | Sensitivity Analysis | Uncertainty Quantification

Education

2023 **Master 2 – MASH** (Mathematics, Machine Learning, Sciences, Humanities) at **Université PSL** (Dauphine & ENS), with Merit (mention Bien).
Research thesis at EDF R&D: *Advanced methods for model calibration with uncertain inputs*.
2022 **Master 1 – Applied Mathematics**, Statistics track, at **Paris Dauphine-PSL**.
2021 **Bachelor's – MEFA** (Mathematics, Economics, Finance, Actuarial Science) at **Paris Dauphine-PSL**
2025 *TOEFL iBT (112/120)*

AI Safety Engagement

Courses ARENA material, self-study, started 14 Oct 2026 — github.com/Edimah/arena
Lens Academy — **"AI Risk Fundamentals"** intensive, certified Oct 2026
BlueDot Impact — **"Future of AI"** course, certified Oct 2026
BlueDot Impact — AGI Strategy, self-paced, from Oct 2026

Reflections A couple of AI safety posts on my personal blog, edimah.github.io; one crossposted to the Effective Altruism Forum

Sprints & hackathons AI Collusion Research Sprint, Apart Research (October 2026)
AI Incident Response Sprint, Apart Research × CeSIA (September 2026)
"Secret Loyalties", Apart Research × Formation Research (July 2026)

Community Slack: EA France, AI Alignment · Discord: AI Safety & SLT, Sûreté de l'IA, Apart, Lens Academy
Paris AI Safety Community ("Nouvelle Machine"): dinners and workshops, participant since Oct 2026 · Effective Altruism France events; registered for EA Summit Paris, 5 Dec 2026

Skills

Data Data management · Data visualisation · Health data regulation · Communication
Statistics Permutation & randomisation tests · Multiple-comparison correction
Change-point detection · Difference-in-differences · Bayesian inference, MCMC
Sensitivity analysis (Sobol) · Uncertainty quantification
Programming Python (PyMC, pytest; PyTorch via ARENA, in progress) · R · SQL · Git, Make, LaTeX
Languages French (native) · English (C2, TOEFL 112/120) · Spanish (B2)